

NEWTON



**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GENERAL CERTIFICATE OF EDUCATION ADVANCED LEVEL**

**PURE MATHEMATICS
PAPER 1**

**6042/1
3 Hours**

JUNE 2026 SESSION

Additional materials:

Answer booklet

Graph paper

List of Formulae MF7

Scientific calculator [Non-Programmable]

8088

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer **all** questions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

This question paper consists of 6 printed pages.

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[Turn over



Answer all questions.

1. Solve the following pair of simultaneous equations.

$$xy = 3$$

$$x - y + 2 = 0.$$

[5]

2. Find the set of values of a for which the line $y = 3x + a$, and the curve $xy = -3$ do not intersect.

[5]

3. Variables x and y are related by the equation

$$ay^2 + by = x^2,$$

where a and b are constants.

When the graph of y against $\frac{x^2}{y}$

is plotted, a straight-line graph is obtained passing through $(2; -5)$ and $(0; -6)$.

Find the value of a and the value of b .

[5]

4. An error of 3% is made in the measurement of the area of a circle.
Find the percentage error in the

(a) radius,

[3]

(b) circumference.

[3]

5. The complex numbers Z_1 and Z_2 are given by $3 - 4i$ and $1 + 2i$, respectively.

(a) Express $\frac{Z_1}{Z_2}$ in the form $a + bi$.

[3]

(b) Find

(i) $\left| \frac{Z_1}{Z_2} \right|$

[2]

(ii) $\arg\left(\frac{Z_1}{Z_2}\right)$

[2]



6. (a) Given that
 $U_n = n^2 - 1$
 write down an expression for
- (i) U_{n-1} [2]
- (ii) $U_{n-1} - U_n$ [2]
- (b) A geometric progression has first term $2\frac{1}{2}$ and common ratio $3\frac{1}{4}$.
 Given that the n^{th} term of the progression is greater than 55,
 find the least number of terms. [4]
7. The position vectors of A, B and C relative to O are
 $3\mathbf{i} + p\mathbf{j} - 3\mathbf{k}$; $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$; $2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, respectively.
 Find
- (a) the value of p for which $\mathbf{OA}$ is perpendicular to $\mathbf{BC}$, [3]
- (b) a unit vector in the direction of the vector $\mathbf{OM}$, where M is the midpoint of $\mathbf{AB}$. [3]
- (c) the acute angle $\mathbf{AOM}$. [3]
8. (a) Express $\sqrt{3} \cos \theta + \sin \theta$, in the form $R \cos (\theta - \alpha)$,
 where $R > 0$ and α is acute. [2]
- (b) Hence, solve the equation
 $3\sqrt{3} \cos 2\theta + 3 \sin 2\theta = 4$, for $0 \leq \theta \leq 360^\circ$. [4]
- (c) Write down the two transformations which map the graph of
 $y = \cos \theta$, onto the graph of
 $y = \sqrt{3} \cos \theta + \sin \theta$. [2]
- (d) Sketch the graph of
 $y = \sqrt{3} \cos \theta + \sin \theta$, for $0^\circ \leq \theta \leq 120^\circ$,
 showing clearly the intercepts. [2]

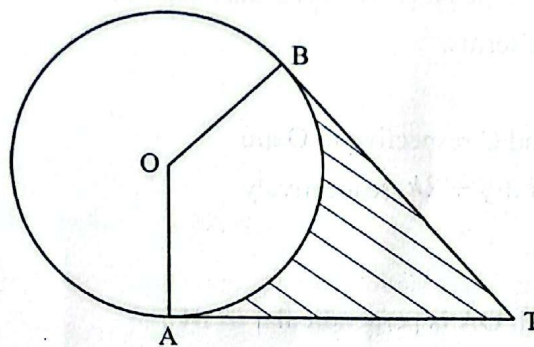


9. The points $A(2;14)$, $B(-2;8)$, C and D are vertices of a rectangle. C lies on the x -axis.

Find the

- (a) equation of the line BC , [3]
- (b) coordinates of C and D , [4]
- (c) area of rectangle $ABCD$. [3]

10.



The diagram shows a circle with centre O and radius 10cm .
 Points A and B lie on the circle.
 The tangents at A and B meet at point T and $AT = BT = 20\text{cm}$.

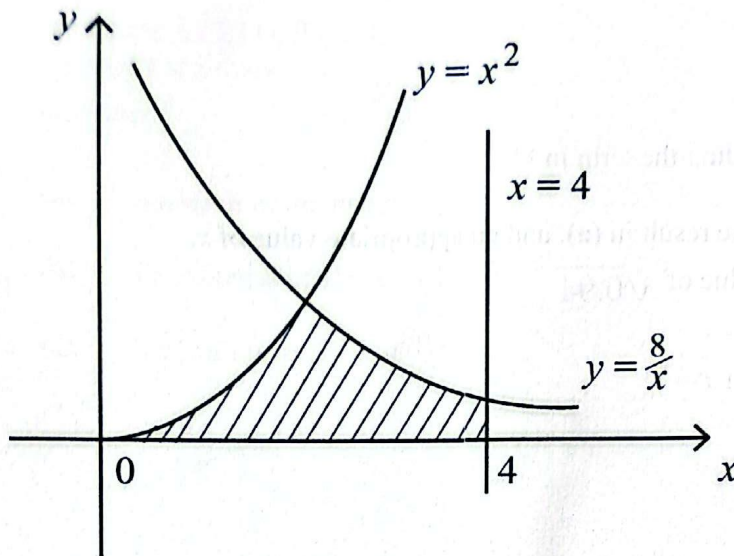
- (a) Show that angle $\hat{AOB} = 2.21$ radians, correct to 3 significant figures. [3]
- (b) Find the
 - (i) perimeter of the shaded region, [3]
 - (ii) area of the shaded region. [4]



11. (a) Expand $(1 - x)^{\frac{1}{2}}$
up to and including the term in x^3 [4]
- (b) Hence, using the result in (a), and an appropriate value of x ,
estimate the value of $\sqrt{0.94}$ [3]
- (c) Integrate $x^3 \ln x$. [3]
12. A curve has equation $y = x^2 - x$.
- (a) Find the set of values for which $y < 6$. [4]
- (b) Express y in the form $a(x + b)^2 + c$
where a, b, c are constants. [2]
- (c) State why y has no inverse. [1]
- (d) Given that y is defined for $x \geq k$,
State the minimum value of k such that y can have an inverse. [1]
- (e) Find the inverse of y and state the domain. [3]



13.



The diagram shows the shaded region enclosed by the graphs

$$y = x^2, y = \frac{8}{x}, y = 0 \text{ and } x = 4.$$

The curve $y = x^2$ and the curve $y = \frac{8}{x}$ intersect at point P.

Find the

- (a) coordinates of point P, [2]
- (b) exact value of the area of the shaded region, [4]
- (c) volume generated when the shaded region is rotated about the line $y = 0$, through 2π radians, leaving the answer in exact form. [5]

14. (a) The equation of a curve is

$$x^3 + y^3 = 9.$$

- (i) Find $\frac{dy}{dx}$ in terms of x and y . [3]
- (ii) Find the equation of the normal to the curve at the point $(1;2)$ [5]

(b) Find x in terms of y given that

$$\frac{dy}{dx} = \frac{1}{x \cos y}$$

$$\text{and } x = 1 \text{ when } y = \frac{3\pi}{2}$$

[5]

